

Assessment Tool 3 – Article Review Answers

Selected Article: Thotad, P., Bharamagoudar, G. R., & Anami, B. S. (2022). *Predictive Analysis of Diabetes Mellitus Using Decision Tree Approach*. 2022 2nd Asian Conference on Innovation in Technology (ASIANCON), IEEE, pp. 1–7. DOI: available via IEEE Xplore, Document ID 9909122. (Submit alongside this review: the IEEE Xplore link <https://ieeexplore.ieee.org/document/9909122/> as the valid record; retrieve the DOI/PDF from your institutional IEEE Xplore access for submission.)

Task 1: Article Summary

(a) Citation & domain Full citation as above. **Domain:** Healthcare / medical diagnostics. **ML problem addressed:** binary classification — predicting whether a patient has Diabetes Mellitus from clinical attributes, using a Decision Tree classifier on the PIMA Indian Diabetes dataset.

(b) Objective & research gap (150–200 words) Diabetes Mellitus is a widespread non-communicable disease that, left undiagnosed, causes serious complications such as nerve damage, heart disease, kidney failure, and stroke. The authors note that global diabetic case counts are projected to rise sharply by 2030, making early, low-cost, and scalable diagnosis a pressing need. Traditional diagnosis relies on clinical blood tests that are time-consuming and resource-intensive, and many existing ML-based screening tools either lack transparency or are not tuned well enough to be clinically trustworthy. The research gap the authors target is the need for a diagnostic model that is both interpretable to clinicians and sufficiently accurate for early screening, without requiring complex infrastructure. Their proposed fix is to build and tune a Decision Tree classifier on lifestyle and physiological risk factors (blood pressure, obesity, age, glucose-related indicators) drawn from the PIMA dataset, arguing that a tree-based model's rule structure gives clinicians an interpretable basis for flagging at-risk patients, unlike opaque black-box alternatives.

Task 2: Methodology Analysis

(a) Technique & dataset The core technique is a **Decision Tree classifier** (a supervised inductive-learning / CO4 topic), trained and evaluated for binary diabetes prediction. **Dataset:** the PIMA Indian Diabetes dataset — a widely used, publicly available benchmark (originally from the National Institute of Diabetes and Digestive and Kidney Diseases) containing several hundred female patient records with features such as glucose concentration, blood pressure, BMI, insulin level, age, and number of pregnancies, labeled as diabetic/non-diabetic. Preparation typically involves cleaning invalid zero-value entries (glucose/BMI cannot legitimately be zero), feature scaling, and a train/test split before fitting the tree.

(b) CO4 mapping & strength/limitation This maps to **Decision Trees** and **Inductive Learning** under CO4 — a rule-based supervised model that generalises from labelled examples. **Strength:** the tree produces a transparent, rule-based structure (e.g., “if Glucose > threshold and BMI > threshold → diabetic”), which is highly interpretable and clinically actionable compared to opaque models. **Limitation:** decision trees are prone to overfitting on a modest-sized dataset like PIMA unless pruning or depth constraints are carefully applied, and a single tree’s performance can be less stable/robust than ensemble alternatives (Random Forest, Gradient Boosting) on the same data.

Task 3: Results & Critical Evaluation

(a) Key results The Decision Tree model reported strong classification performance on the PIMA dataset, consistent with related decision-tree-based diabetes studies in this literature, which commonly report accuracy in the low-to-high 90s% and AUC values in a similar range when tuned well.

Comparative table (typical benchmark figures reported across decision-tree diabetes-prediction studies in this literature, for context):

Model	Accuracy	Precision	Recall/AUC
Decision Tree (this article’s approach)	High 80s–90s%	High	High
Logistic Regression (baseline comparators in related work)	~78–81%	Moderate	Moderate
Naïve Bayes (baseline comparators in related work)	~76–82%	Moderate	Moderate

(b) Critical evaluation The reported metrics are plausible but should be interpreted cautiously: the PIMA dataset is small (~768 records), single-population (Pima Indian women only), and contains known data-quality issues (biologically impossible zero values in several fields), all of which can inflate apparent accuracy and limit generalisability to broader, more diverse patient populations. Potential biases include demographic bias (results may not transfer to male patients or other ethnicities) and evaluation-protocol bias if a single train/test split (rather than k-fold cross-validation) was used, which can overstate performance. Additional experiments that would strengthen the findings: k-fold cross-validation, external validation on a second independent hospital dataset, and reporting confidence intervals around the accuracy/AUC figures.

(c) Real-world applicability Such a model could be deployed as a low-cost pre-screening tool in primary-care or rural clinics with limited diagnostic infrastructure, flagging at-risk patients for confirmatory

blood testing. Scaling challenges include: data availability (larger, more demographically diverse labeled datasets are needed for a production-grade model), compute/integration (embedding the model into existing Electronic Health Record systems), and ethical/regulatory issues (clinical-decision-support tools require regulatory validation, ongoing bias monitoring, and clear human-oversight protocols before affecting real treatment decisions).

Task 4: Reflection & Learning Outcome

(a) Three key insights 1. **Interpretability matters in high-stakes domains** — this article reinforced that in healthcare, a slightly less “powerful” but interpretable Decision Tree can be preferable to an opaque model, directly connecting to the course’s Decision Tree / inductive-learning module. 2. **Data quality drives inductive learning outcomes** — the PIMA dataset’s known data-entry artifacts (invalid zeros) show that inductive learning is only as reliable as its labelled training data, reinforcing the course’s data-preprocessing concepts. 3. **Evaluation protocol shapes reported performance** — comparing this paper’s results with related decision-tree diabetes studies highlighted how choices like train/test split vs. cross-validation materially affect reported accuracy, connecting to the course’s model-evaluation/statistical-learning topics.

(b) Proposed extension Propose extending this work into a **Reinforcement Learning-based longitudinal treatment recommender**: instead of a one-time diagnostic classification, model each patient’s health trajectory as an MDP (as in Task 4 of Assessment Tool 2) where the Decision-Tree diagnosis becomes the initial state, and an RL agent learns which sequential interventions (diet/exercise/medication/monitor) most improve long-term outcomes across multiple patient visits. This is feasible because the same clinical feature set can be reused to define states, and expected impact is a shift from static diagnosis to a decision-support tool that also guides ongoing management, an incremental and clinically valuable extension of the original article’s classification-only scope.

Note: Full paraphrased summary of the article’s findings above is compiled from the publicly available abstract/citation metadata for this IEEE ASIANCON 2022 paper together with comparative figures typical of similar PIMA-dataset decision-tree diabetes studies. For submission, retrieve the exact accuracy/precision/recall/AUC figures and full-text detail directly from the paper via your institution’s IEEE Xplore access (Document ID 9909122) and update the comparative table and quoted numbers in Task 3(a) with the paper’s own reported values before final submission.